

BackTestBench

Test trading ideas on past market data — before risking real money.

- Industrial track · TA: Gerald Imbugwa · 20.07.2026
- Frozen commit: [a2fab591](#)
- A local tool to try, compare, and check strategies safely

Team

Member	Role
Marat Diarov	Team lead — customer communication, frontend, reports
Georgii Beliaev	Tech lead — product design, testing & delivery
Bogdan Zinchenko	Market data & history
Samy Badavi	Trade simulation
Maxim Kharlamov	Results & rankings
Philip Oghenerukevwe Idiare	Strategy rules
Nikolay Taran	Connections to brokers & data sources

Customer & Audience

Our customer: Alexey Potemkin — we met every week (June–July) to show progress and gather feedback.

Who this is for:

- Traders who want to test ideas systematically, not guess from charts alone
- People working with Russian market data (T-Bank and other sources)
- Anyone who needs one **local workspace** — not a paid cloud platform

Problem Statement

Today, testing a trading idea is **scattered and messy**:

- Charts in one app, spreadsheets in another, custom scripts on the side
- Hard to **compare** strategies or settings side by side
- No single place to **check** whether good past results still hold on other dates
- We need a **safe sandbox** — research only, not real-money trading claims

Our Solution

BackTestBench — one place for the full research flow:

1. **Load history** — price data from brokers you connect
2. **Set your rules** — when to buy, sell, take profit, stop loss
3. **Run a test** — see profit/loss and risk on past data (virtual trades)
4. **Compare results** — rank strategies, health verdicts, and top parameter sets

Runs locally in one package — same setup for the whole team.

Key Features

What you need	What you get
Market data	History from T-Bank, TwelveData, Bybit, Binance; saved for faster reruns
Strategies	Flexible rules you can edit — profit targets, stops, filters
Testing	Run one setup or try many parameter combinations automatically
Results	Profit/loss, risk metrics, charts with buy/sell points
Dashboard	One web screen: run backtests, rank strategies, tune parameters

How You Use It

1. Open the **web dashboard** in your browser
2. Pick an instrument and a strategy; adjust settings if needed
3. Click **Run** — wait for the chart and numbers to update
4. **Scroll strategy cards** — rank badges, metrics, charts, parameter search, health verdict

Built with Python and a web interface · runs locally via Docker · English dashboard ·
MOEX/Russian-market data sources

What Happens (Simple)

Four steps inside the product — no coding required from the user:

1. **Get market data** from your connected source
2. **Apply your strategy rules** candle by candle
3. **Simulate trades** — what would have happened in the past
4. **Show results** — metrics, chart, strategy ranking, and top parameter sets

Optional **Live refresh** re-runs one strategy on new candles — still simulated, not live trading.

Measured Improvement

	Early prototype (Weeks 3–4)	Final submission (a2fab591)
Automated tests	~66	206 (78% of core code covered)
Quality on every change	Basic smoke test	Full automated pipeline on each pull request
What the user can do	One demo strategy	Edit rules, search parameters, multiple data sources, ranking & health

Value to Customer

What changed for Alexey:

- Started with **one fixed demo strategy** → now a **full research workspace**
- Can search parameters and discuss risk with real numbers, not hand-waving
- July feedback focused the roadmap on **reliability of settings** and **stronger date-range tests** — not more flashy demos
- Everything in **one screen**: run backtest → compare ranked cards → discuss numbers

Challenges & Lessons Learned

- **Flexible rules are powerful but harder to get right** — some preset settings still need fixes
- **Fast first version** — great for demos, but run history is not permanently stored yet
- **Customer meetings kept us focused** — quality and validation over feature count
- **Simulation only** — the app does not place real broker orders; that is intentional
- **Multi-period date-range checks** — not in the dashboard yet; customer wants this next

Future Work

Next priorities (from customer, July 13):

1. Make sure every strategy preset applies correctly when you run a test
2. Stronger checks across **multiple time periods** in rankings
3. Show **percentage and risk/reward** metrics, not only absolute profit

Later: saved run history · more instruments in the UI · richer data quality checks

Not our goal: live trading · promising future profits

Live demonstration
